Q1. Describe the ateiology, pathology, clinical features and management of actinomycosis.

Q2. Discuss etiology, pathology, clinical features and management of tetanus.

Q3. Define carbuncle. Mention its ateiology, clinical features and treatment.

Q4. Discuss pathogenesis, clinical features and management of gas gangrene.

Q5. Describe the signs, symptoms and treatment of alveolar abscess.

Q6. Ludwig's Angina.

Q7. Erysipelas

Q8. Boils

Q9. Syphilis

Q10. Precautions for Surgeon-HIV-Infected Patient or

Universal precautions

Q11. Hyperkalaemia

Q12. Chronic Osteomyelitis of the Jaw

Q13. Cancrum Oris (Noma)

Q14. AIDS (Acquired Immunodeficiency Syndrome). Discuss briefly spread and prevention of AIDS

Q15. Diagnostic and Confirmatory Tests in AIDS

Q1. Describe the ateiology, pathology, clinical features and management of actinomycosis.

Actinomycosis

Aetiology

- Chronic bacterial infection caused by **Actinomycetes** (filamentous, gram-positive, non-acid-fast, non-spore-forming, anaerobic bacteria).

Pathophysiology

- Requires a break in **mucosa integrity** + **devitalized tissue** for infection.
- **Polymicrobial infection** – needs companion bacteria to produce toxins or suppress host defenses.
- Host response: **intense inflammation → fibrosis**.
- **Spreads contiguously**, ignoring tissue planes.
- Leads to **draining sinus tracts**; rare **lymphatic spread** but hematogenous spread possible.

Clinical Symptoms

A. Cervicofacial Actinomycosis

- History: **Dental trauma**, poor hygiene, **dental caries**, or jaw damage (neoplasm, osteonecrosis).
- **Painless/painful swelling** in the submandibular/perimandibular region → forms **sinuses** with **sulfur granules** in exudate.
- **Skin discoloration**: reddish/blue.
- **Nodules**: Tender early → **woody hard** later.
- No lymphadenopathy.
- **Trismus** if mastication muscles involved.
- Variable **fever**, difficulty chewing.

B. Thoracic Actinomycosis

- History: **Aspiration** (risk: seizures, alcoholism, poor hygiene).
- Symptoms: **Cough** (dry/productive), **blood-streaked sputum**, chest pain, shortness of breath.
- **Sinus tracts** from chest wall (**pleurocutaneous fistula**).
- **Systemic**: Fever, weight loss, fatigue, anorexia.

C. Abdominal Actinomycosis

- History: Abdominal surgery, perforation, or foreign body ingestion.
- Symptoms: **Sinus tracts** (drainage from abdominal wall or perianal region).

- **Nonspecific symptoms**: Low-grade fever, weight loss, fatigue, nausea, vomiting, abdominal discomfort, bowel habit changes.

D. Pelvic Actinomycosis

- History: **IUCD use**.
- Symptoms: **Lower abdominal pain**, vaginal bleeding/discharge.

Management

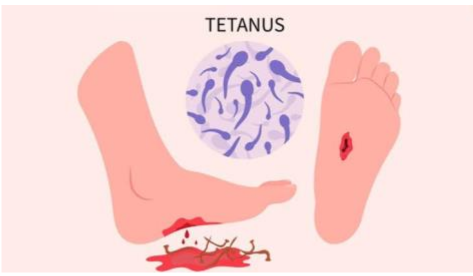
A. Diagnosis

1. **CBC**: Mild **anemia**, **leucocytosis**.
2. **ESR, CRP**: Elevated.
3. **Culture**:
    - Identify organism from **sulfur granules** or clinical specimen.
    - **Gram stain**: Beaded, branched, gram-positive rods.
4. **Imaging**: Chest X-ray, CT, USG-guided aspiration, biopsy, exploratory surgery if needed.

B. Treatment

1. **Medications**:
    - **Penicillin G**: Drug of choice.
    - Alternatives: **Doxycycline** (penicillin allergy) or **Sulfonamides** (e.g., sulfamethoxazole 2–4 g/day).
    - Slow response, requires months of treatment.
2. **Surgery**:
    - **Drain abscesses**, remove sinus tracts, excise fibrotic lesions.
    - Relieve obstruction if present.

Q2. Discuss etiology, pathology, clinical features and management of tetanus.



Tetanus

- **Definition:** Emergency condition with **prolonged skeletal muscle contraction.**

Aetiology & Pathology

- **Cause: Clostridium tetani** (anaerobic gram-positive bacillus).
- **Spores**: Found in **soil, dust, intestines**; resist heat/disinfectants.
- **Pathogenesis**: Spores enter tissue → germinate in **anaerobic conditions** → release toxins:
  - **Tetanospasmin**: Blocks neurotransmitters → **muscle spasms**.
  - **Tetanolysin**: Role unclear.

Clinical Features

- Incubation: 4–14 days.
- **Symptoms:**
  - **Trismus (lockjaw):** 75% cases.
  - **Stiffness**: Starts in **jaw/neck** → spreads in 24–48 hrs.
  - **Spasms**: Triggered by noise/light → risk of **apnoea, fractures, asphyxia.**
  - **Risus sardonicus:** Sneering grin.
  - Severe: **Opisthotonus**, diaphragm spasms → **autonomic dysfunction.**

Diagnostic Test

- **Spatula test:** Reflex spasm → patient bites spatula.

Types

- Generalized: Most common; starts with trismus → spreads.
- **Neonatal**: Due to infected **umbilical stump.**
- **Local**: Spasms near injury site.
- **Cephalic**: Involves **cranial nerves.**

Management

1. Diagnosis:

- **Clinical signs:** Trismus, spasms.
- Labs: CBC, ESR, CRP (nonspecific).

2. Wound Care:

- **Clean + debride necrotic tissue.**

3. Antibiotics:

- **Metronidazole** IV (10 days).

4. Passive Immunization:

- **TIG (3000–5000 units IM)** + TT vaccine.

5. Supportive Care:

- **Diazepam**: Spasms.
- **ICU**: For severe cases; intubation/tracheostomy if needed.
- **Nutrition**: High-calorie, high-protein diet.
- Recovery: 4–6 weeks.

Q3. Define carbuncle. Mention its ateiology, clinical features and

treatment.

Definition

- **Carbuncle**: A **skin infection** involving a group of hair follicles.
- It is a **confluence of multiple furuncles** (skin boils) forming a larger abscess.
- Contains **pus** and may have multiple **openings draining onto the skin**.

Aetiology

- Caused by **Staphylococcus aureus** (most common).
- Contagious: Can spread to other body areas or people.
- Risk factors:
  - **Poor hygiene**, friction (e.g., clothing/shaving).
  - **Weakened immunity**, **diabetes**, or dermatitis.

Clinical Features

- **Cluster of boils** filled with **pus, fluid, and dead tissue**.
- Common sites: **Back**, **nape of neck**.
- **Men** are more prone than women.
- Size varies: From a **pea** to a **golf ball**.
- Symptoms:
    - **Red, painful swelling** with a **white/yellow center**.
    - May **crust or spread** to other areas.
    - **Fatigue, fever**, and general discomfort may occur.
    - **Itching** may precede its appearance.

Treatment

Self-Care

- **Warm, moist compress**: Helps drain and heal.
- Avoid squeezing or cutting without medical supervision.

Medical Treatment

- For carbuncles >2 weeks old, recurrent, with fever, or on critical locations:
    - **Antibacterial soap** for cleansing.
    - **Antibiotics** for infection control.
    - **Surgical drainage**:
        - Deep lesions need **cruciate incision** under sterile conditions.

Prevention of Spread

- Maintain **proper hygiene**.
- Wash hands with **antibacterial soap** after touching the carbuncle.
- Avoid sharing towels, clothes, or sheets.
- Wash infected items in **boiling water**.
- Change bandages frequently and dispose of them properly.
- Use daily **antibacterial soaps** to suppress bacteria if recurrence is frequent.

Q4. Discuss pathogenesis, clinical features and management of gas gangrene.

(**Clostridial Myonecrosis**)

Definition

- **Bacterial infection** of muscle tissue caused by **toxin-producing clostridia**.
- Deadly form of gangrene, usually caused by **Clostridium perfringens**.
- It is a **medical emergency**.

Pathogenesis

- Caused by **anaerobic, gram-positive, spore-forming bacilli** (e.g., **C. perfringens**).
- Other species: **C. septicum, C. novyi, C. histolyticum**, etc.
- **Alpha-toxin**: Main toxin with **lecithinase (phospholipase-C)** activity, causing:
    - **Cell destruction** (lyses RBCs, WBCs, platelets, muscle cells).
    - **Necrosis**, **oedema**, and **blood vessel thrombosis**.
- **Gas production**: From glucose fermentation in infected tissues.
- Systemic effects:
    - **Severe hemolysis** → Low hemoglobin → **Renal failure**.
    - Progression to **shock** if untreated.

Clinical Features

- Often follows **trauma**, **surgery**, or **occult malignancy**.
- Symptoms:
    1. **Sudden severe pain**, spreading with the infection.
    2. **Swelling**, **serosanguineous exudate**.
    3. Skin turns **bronze → blue-black**, with **bullae** and **blebs**.
    4. **Crepitus** (gas under skin), but may be absent due to oedema.
    5. Wound may have **sweet/mousy odor** or no odor.
    6. **Late signs**:
        - **Hypotension**, **renal failure**, and **mental clarity paradox**.
        - Fever, **tachycardia**, and altered mental state.

Management

1. Surgical Treatment

- **Debridement**: Aggressively remove all necrotic tissue.
- **Amputation**: If necessary to control infection.
- **Fasciotomy**: For compartment syndrome.

2. Antibiotic Therapy

- **Penicillin G + Clindamycin**: Preferred combination.
- Alternatives for penicillin-allergic patients:
    - **Clindamycin + Metronidazole**.

3. Hyperbaric Oxygen Therapy (HBO)

- Increases oxygen supply to tissues, aiding in toxin neutralization.
- Used as an adjunct to surgery and antibiotics.

Summary of Treatment

- **Early diagnosis** and prompt **surgical intervention** are critical.
- **Antibiotics**: Penicillin + Clindamycin or alternatives.
- **HBO therapy**: Adjunctive but lacks strong clinical evidence.

Q5. Describe the signs, symptoms and treatment of alveolar abscess.

Alveolar Abscess (Dentoalveolar Abscess)

Definition

- **Abscess** in the **alveolar ridge** of the jaw caused by infection from an adjacent **nonvital tooth**.
- Localized **suppurative inflammation** around the root apex of a tooth.

Pathology

1. **Periapical Abscess**:
   - Originates in the **dental pulp** (common in children).
   - Progression: **Dental caries → Pulpitis → Necrosis → Abscess**.
2. **Periodontal Abscess**:
   - Involves **periodontal ligaments** and **alveolar bone** (common in adults).
3. **Pericoronitis**:
   - Infection of **gum flap** over a **partially erupted or impacted third molar**.

Clinical Features

1. **Local Symptoms**:
   - **Pain**: Localized and progressive.
   - **Swelling**: Develops over hours/days.
   - **Thermal sensitivity**: Due to dentin exposure.
   - **Gingival bleeding** (sometimes in periodontal abscess).
   - Reduced intake of food and fluids.
2. **Physical Signs**:

   i. **Gingiva**:
   - Swelling, **warmth**, **erythema**, and fluctuance (towards buccal side).
   - **Parulis ("gum boil")**: Reddish papule at the endpoint of a draining sinus tract.

   ii. **Teeth**:

   - Commonly involves **lower third molar**, followed by other posterior teeth.
   - **Tenderness** on pressure/percussion, mobility, or extrusion.

   iii. **Regional Lymph Nodes**: May be involved.

   iv. **Severe Infection**:

   - **Trismus** (indicates masticator space involvement).
   - **Dysphagia** (difficulty swallowing).
   - **Respiratory difficulty**, **necrotizing fasciitis**, **neck/facial swelling**.

Treatment

1. Airway Management

- Assess for respiratory distress or oropharyngeal swelling.
- Secure airway with **intubation** or **tracheostomy** if needed.

2. Diagnosis

- Collect **specimens** for Gram stain and **culture** (aerobic/anaerobic).

3. Medications

- **Antibiotics**: Empiric therapy to control infection.
- **Analgesics**: Pain relief.
- Ensure **hydration**.

4. Procedures

- **Pulpectomy** or **incision and drainage**: For localized acute apical abscess.
- **Tooth extraction**: Common surgical option for the affected tooth.

Q6. Ludwig's Angina.

Ludwig's Angina

Definition:

- A **life-threatening cellulitis** of the **submental and submandibular regions,** with **inflammatory edema** of the mouth.
- Common in **adults** with **dental infections**, caused by **streptococcal organisms.**

Causes:

- **Dental infections** (second/third molars).
- **Poor oral hygiene** or periodontal disease.
- **Trauma** to the mouth.
- Systemic conditions like **diabetes** or **immunosuppression.**

Clinical Features:

- **Swelling:**
  - **Diffuse swelling** in the submandibular region.
  - **Tongue pushed upward,** causing **swallowing difficulty.**
- **Symptoms:**
  - **Putrid halitosis, fever,** and **toxicity.**
  - **Wooden induration** in the submandibular area.
- **Other Manifestations:**
  - **Trismus (**lockjaw**), hoarseness, and dyspnea** from glottic edema.

Complications:

- **Airway obstruction, septicemia, mediastinitis,** or **aspiration pneumonia.**

Treatment:

- **Conservative Management:**
  - **Hospitalization** with **IV antibiotics** (ampicillin-sulbactam, clindamycin).
  - **Ryle's tube feeding** for swallowing issues.
  - **IV fluids** for hydration.
- **Surgical Intervention:**
  - **Incision and drainage: A 5-6 cm incision** in the submandibular region to drain pus.
  - **Irrigate** the cavity and use **loose sutures** for drainage.
- **Airway Management:**
  - **Tracheostomy** if severe obstruction occurs.

Prevention:

- Maintain **oral hygiene** and treat dental infections early.
- Manage systemic diseases like **diabetes**.

Key Points:

- Ludwig's Angina is a **medical emergency** requiring **prompt treatment** to avoid airway obstruction and complications.

Q7. Erysipelas

Definition

- **Erysipelas** is an **acute superficial Streptococcus bacterial infection** affecting the **deep epidermis**, characterized by **cutaneous lymphatic spread**.

Aetiology and Pathology

1. **Causes**:
    - Most cases are caused by **Streptococcus pyogenes** (also known as **beta-hemolytic group A streptococci**).
    - Non-group A streptococci can also be responsible.
2. **Development**:
    - Infection typically starts with **bacterial inoculation** into skin trauma.
    - **Local factors** that increase risk:
        - **Venous insufficiency**, **stasis ulcerations**, **inflammatory dermatoses**, **dermatophyte infections**, **insect bites**, and **surgical incisions**.
    - The infection rapidly invades and spreads through **lymphatic vessels**, causing **skin "streaking"** and regional **lymph node swelling**.

Clinical Features

1. **General Symptoms**:
    - High **fever**, **shaking chills**, **fatigue**, **headaches**, **vomiting**, and a general feeling of illness within **48 hours** of infection.
2. **Skin Symptoms**:
    - Initially, **erythematous patch** (redness) that evolves into a **fiery-red**, **indurated**, **tense**, and **shiny plaque**.
    - The lesion enlarges quickly with **sharply demarcated raised edges**.
    - The skin is **red**, **swollen**, **warm**, **hardened**, and **painful**, resembling the consistency of **orange peel**.

    - Severe cases may develop **vesicles**, **bullae**, and **petechiae**, possibly leading to **skin necrosis**.
3. **Lymphatic Involvement**:
    - **Skin streaking** and **regional lymphadenopathy** (swollen, tender lymph nodes) are common signs of lymphatic involvement.
4. **Common Locations**:
    - The infection can affect any part of the skin, including the **face**, **arms**, **fingers**, **legs**, and **toes**.
    - It commonly affects the **extremities**, with **fat tissue** being more susceptible.
    - **Facial involvement** often occurs around the **eyes**, **ears**, and **cheeks**.

Treatment

1. **Supportive Care**:
    - **Elevation** and **rest** of the affected limb to reduce **swelling**, **inflammation**, and **pain**.
    - **Saline wet dressings** should be applied to ulcerated or necrotic lesions and changed every **2-12 hours**, depending on infection severity.
2. **Medications**:
    - **Penicillin** is the **first-line therapy** for erysipelas.
        - Administer **oral or intramuscular penicillin** for **10-20 days** for most classic cases.

Q8. Boils

Definition

- A **boil** (or **furuncle**) is a **deep folliculitis**—an infection of the **hair follicle**.
- Most commonly caused by **Staphylococcus aureus**.
- Results in a **painful, swollen area** on the skin due to the **accumulation of pus** and dead tissue.
- When multiple boils form together, it is called a **carbuncle**.

Clinical Features

1. **Appearance**:

    - **Bumpy**, **red**, **pus-filled lumps** around a hair follicle.
    - **Tender**, **warm**, and **very painful** to the touch.
    - Size ranges from **pea-sized** to **golf ball-sized**.
    - A **yellow or white point** at the center of the lump indicates it is ready to **drain** or discharge **pus**.
2. **Systemic Symptoms**:
    - In severe infections, **fever**, **swollen lymph nodes**, and **fatigue** may occur.
3. **Risk Factors**:
    - **Systemic factors** that lower immunity and increase susceptibility:
        - **Diabetes**, **obesity**, and **hematologic disorders** (e.g., blood-related conditions).

Treatment

1. **Home Care**:
    - **Warm salt water compresses** can help encourage **draining**.
    - Wash and cover the furuncle with **antibiotic cream** or **antiseptic tea tree oil**, then apply a **bandage** to promote healing.
2. **Medical Supervision**:
    - **Never squeeze** or **lance** a furuncle without a **doctor's oversight** to avoid spreading the infection.
3. **Antibiotics**:
    - **Antibiotic therapy** is recommended for:
        - **Large** or **recurrent boils**.
        - Boils occurring in **sensitive areas**.

Q9. Syphilis

Definition

- **Syphilis** is a **venereal disease** caused by the spirochete **Treponema pallidum**.

- Transmitted through **sexual contact**, **mother to fetus** (in utero), **blood transfusions**, or through **breaks in the skin** coming into contact with infectious lesions.

Stages of Syphilis

1. **Primary Syphilis**
   - Occurs **3 weeks** after contact with an infected person.
   - Presents as a **solitary, raised, firm papule** on the **glans penis** (in males) or **vulva/cervix** (in females).
   - The lesion (chancre) **ulcerates** into a **crater** with **elevated edges**.
   - Heals **in 4-8 weeks**, with or without treatment.
2. **Secondary Syphilis**
   - Occurs **2-10 weeks** after primary chancre.
   - Presents as a **skin rash** (often **non-itchy**, **symmetrical**, on **palms** and **soles**).
   - May also show **mucous membrane lesions** (condyloma latum) and **generalized, non-tender lymphadenopathy**.
   - The rash may evolve into **maculopapular** or **pustular** forms.
3. **Latent Syphilis**
   - The disease is **asymptomatic** during this phase.
   - Can last **from a few years to 25 years**.
   - Diagnosis requires **serologic tests** as there are no symptoms.
4. **Tertiary Syphilis**
   - A **slowly progressive** stage that may affect **any organ**.
   - Lesions (e.g., **gumma**) appear after **3-10 years**.
   - **Bone pain** (described as **deep and worse at night**) is common.
   - Can involve the **CNS**, presenting with symptoms depending on the affected area.
   - Not usually **infectious** at this stage.

Diagnosis

1. **Blood Tests**
   - **Nontreponemal tests**: **VDRL** and **rapid plasma regain**.
   - **Treponemal tests** for confirmation:
     - **Treponema pallidum particle agglutination (TPHA)**
     - **Fluorescent treponemal antibody absorption test (FTA-Abs)**.

Treatment

1. **Penicillin** is the mainstay of treatment.
   - **Primary/Secondary Syphilis**:

        - **Benzathine penicillin G**: 2.4 million units **IM** in a **single dose**.
    - **Early Latent Syphilis**:
        - Same as above: **2.4 million units IM** in a **single dose**.
    - **Late Latent/Unknown Duration Latent Syphilis**:
        - **7.2 million units** in total, given as **3 doses** of **2.4 million units IM** at **1-week intervals**.
    - **Pregnancy**: Treatment varies according to the **stage of syphilis**.

Q10. Precautions for Surgeon-HIV-Infected Patient

- **HIV** (Human Immunodeficiency Virus) causes **AIDS** (Acquired Immunodeficiency Syndrome).
- Special precautions must be followed by surgeons while treating **HIV/AIDS** patients to prevent transmission.

Precautions in OPD

1. **Gloves**:
    - Always wear **gloves** when examining patients with **open wounds**.
2. **Disposable Instruments**:
    - Prefer using **disposable instruments** to minimize the risk of cross-contamination.
3. **Reusable Instruments**:
    - Clean **reusable instruments** with **soap and water**.
    - Immerse them in **glutaraldehyde** (a disinfectant) for proper sterilization.

Precautions in Operating Room

1. **Dental Chair**:
    - Cover the **dental chair** with a **single sheet** of **polythene** to maintain hygiene.
2. **Personnel**:
    - Keep the **number of personnel** in the operating room to a **minimum**.
    - **Staff with abrasions or lacerations** on hands should not enter the operating room.
    - All staff should wear **over-shoes**, **gloves**, **disposable water-resistant gowns**, and **eye protectors**.
3. **Surgical Technique**:
    - **Avoid sharp injuries** during surgery.
    - Take special care to **avoid needle-stick injuries**.
4. **Post-Surgery**:
    - Ensure **proper autoclaving** of all instruments after surgery to sterilize them thoroughly.

Post-Exposure Prophylaxis for Health Workers

- In case of **exposure** to infected material from an **AIDS patient**, health workers should receive medication such as:
  - **AZT (Zidovudine)**
  - **Lamivudine**
  - **Indinavir**

Q11. Hyperkalaemia

Causes of Hyperkalaemia

1. **Impaired Excretion**:
   - **Acute renal failure**
   - **Severe chronic renal failure (CRF)**
   - **Addison's disease**
   - **Hypoaldosteronism**
   - **Use of potassium-sparing diuretics**
2. **Excessive Intake**:
   - **Intravenous fluids** containing **K+**
   - **High K+ foods**
3. **Tissue Breakdown**:
   - **Haemolysis** (destruction of red blood cells)
   - **Rhabdomyolysis** (muscle breakdown)
   - **Crush injury**
4. **Shift of K+ Out of Cell**:
   - **Acidosis** (increase in blood acidity)
   - **Insulin deficiency**
5. **Pseudohyperkalaemia**:
   - **Haemolysed blood sample**

Clinical Features

- **Cardiac arrhythmias**
- **Muscular weakness** progressing to **flaccid paralysis**
- **Respiratory embarrassment** (difficulty breathing)

Electrocardiographic manifestations of hyperkalaemia:

- **Tall peaked T waves**
- **Prolongation of PR interval**
- **Reduced height of P wave**
- **Prolongation of QRS complex**

- **Sine wave pattern**
- **Ventricular fibrillation** and **standstill** may occur terminally.

Management

1. **Identification and elimination** of the underlying cause.
2. **If marked ECG changes**, give:
    - **10 ml of 10% calcium gluconate solution** intravenously, slowly over **5-10 minutes**.
3. **Shift of potassium into cells**:
    - **Intravenous glucose with insulin**:
        - **50 ml of 50% glucose** + **10 units of soluble insulin** intravenously as a **bolus**.
        - Alternatively, **500 ml of 20% glucose** + **10 units soluble insulin** as an **infusion** over **6-12 hours**.
4. **Correct acidosis**:
    - **50-100 ml of 8.4% sodium bicarbonate** intravenously.
5. **Nebulisation of beta-agonists**:
    - **Salbutamol** or **terbutaline** to reduce potassium levels.
6. **Cation exchange resins**:
    - **Sodium polystyrene sulphonate** (oral or enema) helps remove potassium.
7. **If above measures fail**, **haemodialysis** may be needed.

Q12.Describe Chronic Osteomyelitis of the Jaw

It is the **persistent abcess of the bone** characterized by the **complex inflammatory process** including **necrosis of mineralized and marrow tissues, suppuration, resoption, sclerosis and hyperplasia .**

Pathogenesis

- **Infection of bone marrow** from infected pulp.
- **Extension of infection** into cancellous bone.
- **Thrombus formation** in nutrient vessels of the bone.

- **Death of cancellous bony trabeculae**, leading to formation of **sequestrum**.
- **Spread of infection** via **Volkmann's canal** in cortical plates.
- **Periostitis** (inflammation of the periosteum).
- **Multiple sinus tract formation**.
- **Necrosis of cortical bone**.
- **Discharge of pus** through **sinuses** (called **clocae**).

Investigations:

- **Clinical Examination:** Tenderness, swelling, and redness over the affected jaw.
- **X-ray**: Helps to detect bone destruction and **radiolucent areas** indicating infection.
- **CT Scan/MRI:** Provides detailed imaging of bone and soft tissue involvement.
- **Blood Tests:**
    - Elevated **WBC count.**
    - Raised **ESR** and **CRP** indicating inflammation.

Microbiological Culture: For identification of causative organism from pus or blood.

Treatment:

- **Antibiotics:**
    - Empirical broad-spectrum antibiotics until culture results are available.
    - Once the organism is identified, use specific antibiotics (e.g., **penicillin, clindamycin).**
- **Surgical Drainage:**
    - If abscess or pus formation is present, surgical drainage is necessary to remove infected material.

- **Pain Management:**

    - Use of **analgesics** like ibuprofen or paracetamol.

- **Supportive Care:**

    - Hydration and nutritional support.

- **Root Canal Treatment/Extraction**: If the infection is related to a tooth, root canal treatment or tooth extraction may be necessary.

- **Hyperbaric Oxygen Therapy**: In severe or chronic cases, to promote healing and enhance antibiotic effectiveness.

Clinical Features

- **Molar area of mandible** is more commonly affected.
- **Pain** is usually mild and insidious, not related to disease severity.
- **Jaw swelling** is common, but **mobility of teeth** and **sinus tract formation** are rare.
- **Anaesthesia** and **paraesthesia** of lip are uncommon.
- **Regional lymphadenopathy** is common.
- **Thickened, wooden feeling** of bone with slow increase in jaw size.

Histopathology

- **Chronic inflammatory reaction** of bone with **accumulation of exudate** and **pus** within medullary spaces.
- Inflammatory cells: **Lymphocytes**, **macrophages**, and **plasma cells** predominate.
- **Osteoblastic and osteoclastic cavities** partially form with **irregular bony trabeculae** having **reversal lines**.
- **Sequestrum** may develop in later stages.
- **Colonies of bacteria** can be seen within the inflamed tissue.

Radiological Findings

- **Moth-eaten appearance** with **irregular, enlarged radiolucent areas** separated by islands of normal bone.
- **Sequestra**: Irregular calcified areas separated from normal bone.
- **External resorption** of tooth roots and **lamina dura** becomes less apparent.
- **Fistulous tracts** may appear.

Q13. Cancrum Oris (Noma)

1. **Cancrum Oris is always an extension of acute ulcerative gingivitis (Vincent's disease)**
    - **Vincent's disease** is an acute, necrotizing infection of the gums.

    - **Cancrum oris** begins as a complication of this gingivitis, where the infection spreads into the adjacent **soft tissues** like the cheeks, lips, and jaw.
    - The infection causes rapid destruction of **soft tissue** and **bone** in the affected area.
2. **Bone exposed by the necrosis of soft tissues usually sequestrates in about 3 weeks**
    - As the infection progresses, it causes **necrosis** (tissue death) of the soft tissues.
    - This **necrosis** leads to the exposure of **underlying bone**, which becomes infected.
    - In about **3 weeks**, the infected bone becomes **sequestered**, meaning the dead bone is separated from the healthy tissue.
    - This sequestration can complicate healing and necessitate **surgical intervention** to remove the necrotic bone.
3. **After removal of loose bone and antibiotic therapy, considerable scarring occurs, producing facial asymmetry and trismus**
    - Once the infected bone is removed, and **antibiotics** are administered to control the infection, the area heals but with significant **scarring**.
    - **Scarring** can lead to **facial asymmetry**, where the face appears uneven due to tissue loss or abnormal healing.
    - **Trismus**, or restricted mouth opening, can also occur as the infection causes stiffness and scarring of the jaw muscles and tissues, making it difficult to fully open the mouth.

Q14. AIDS (Acquired Immunodeficiency Syndrome). Discuss briefly spread and prevention of AIDS

AIDS is the end stage of a progressive state of **immunodeficiency**, caused by the **Human Immunodeficiency Virus (HIV)**.

Causative Organism:

- **Human Immunodeficiency Virus (HIV)**

Mode of Transmission:

1. **Sexual intercourse** (unprotected)
2. **Mother to fetus** (during pregnancy, childbirth, or breastfeeding)
3. **Contaminated needles** (commonly among drug abusers)
4. **Contaminated blood transfusion**

Clinical Features:

- **Prolonged diarrhoea**

- **Weight loss**
- **Night sweating**
- **Prolonged fever**

Oral Manifestations of AIDS:

1. **Candidiasis (oral thrush)**
2. **Angular cheilitis** (inflammation and cracking at the corners of the mouth)
3. **Kaposi's sarcoma** (a type of cancer that forms in the skin and mucous membranes)
4. **Hairy leukoplakia** (white patches on the tongue)
5. **Herpes simplex and herpes zoster infections**
6. **Ulcerations** in the mouth and throat
7. **Delayed wound healing** (due to the weakened immune system)

Treatment:

- **AIDS** is **incurable**, but certain drugs are used to **prevent disease progression** and manage symptoms.
- **Antiviral drugs**, like **azithromidine** and **zidovudine**, are the most effective treatments against **HIV**.
    - **Zidovudine** may help prolong the life of **AIDS** patients.
    - These drugs work by inhibiting the enzyme **reverse transcriptase**, which prevents the replication of the virus.

Prevention and Control:

1. Safe Sexual Contact:

- **Use of condoms**: Always use a condom during sexual intercourse to reduce the risk of HIV transmission. This is the most effective way to prevent the sexual spread of HIV.

2. Prevent Sharing of Needles:

- **Avoid sharing needles**: Ensure that each person uses their own **sterilized needle** to prevent the spread of HIV, especially among drug abusers.
- **Sterilized needles**: If needles must be used, they should be properly sterilized to prevent contamination.

3. Safe Blood Transfusion:

- **HIV testing before transfusion**: All **blood donations** should be tested for HIV before being transfused to patients. This prevents the accidental transmission of HIV through contaminated blood.

4. HIV Testing and Counseling:

- **Regular HIV testing**: Encourage regular HIV screening for people at high risk, such as those with multiple sexual partners, people using intravenous drugs, or individuals who have been exposed to known HIV-positive individuals.
- **Counseling and education**: Educate at-risk populations about the importance of safe sexual practices, needle safety, and regular HIV testing.

5. Prevention of Mother-to-Child Transmission:

- **Antiretroviral therapy (ART)** for pregnant women with HIV: Pregnant women with HIV should receive **ART** to lower the chance of passing the virus to their unborn child.
- **C-section delivery**: In some cases, a **C-section** delivery may be recommended to reduce the risk of transmission during childbirth.
- **Breastfeeding alternatives**: HIV-positive mothers are advised not to breastfeed their babies, as the virus can be transmitted through breast milk.

6. Post-Exposure Prophylaxis (PEP):

- **PEP treatment**: If an individual is exposed to HIV, they should start **post-exposure prophylaxis (PEP)** treatment within **72 hours** to reduce the risk of infection. This involves taking a combination of HIV medicines for 28 days.

7. PrEP (Pre-exposure Prophylaxis):

- **PrEP**: For high-risk individuals, **PrEP** (a daily pill containing two HIV medicines) can be taken to prevent HIV infection. It is recommended for individuals who are at high risk of HIV exposure.

Q15. Diagnostic and Confirmatory Tests in AIDS

Tests for HIV:

a) The ELISA (Enzyme-Linked Immunosorbent Assay)

- **Standard screening test** for HIV.
- A **blood test** that detects HIV antibodies.
- Requires **two visits**:
    1. **Pre-test counselling** and blood sample collection.
    2. **Post-test counselling** and medical referrals if the test is positive.

b) Western Blot

- **Antibody detection test**, similar to ELISA.
- **Viral proteins are separated** and immobilized.
- The **binding of serum antibodies** to specific HIV proteins is visualized to confirm HIV infection.

c) Rapid HIV Testing

- Provides **pre-test and post-test counselling**, results, and referrals **in one visit**.
- **Fast results** in a short time.
    1. **OraQuick/OraQuick Advanced Rapid HIV-1/2 Antibody Test**
        - Detects **HIV-1 and HIV-2** in **venous blood**, **plasma**, or **oral fluids**.
        - **Approved for use** in various fluids.
    2. **Reveal G2 HIV-1 Antibody Test**
        - Approved for **plasma or serum** specimens.
        - Takes **3 minutes** to develop.
        - Requires **centrifuged serum or plasma**.
        - A **red dot** appears if HIV is present.
    3. **Uni-Gold Recombigen HIV-1 Test**
        - Approved for **whole blood**, **plasma**, or **serum**.
        - Can be used for **venipuncture** or **finger stick**.
    4. **Multispot HIV-1/HIV-2 Rapid Test**
        - Approved for **fresh or frozen plasma**, **whole blood**, or **serum**.
        - Consists of a **test cartridge** and **five reagents** for detecting HIV-1/HIV-2.

Human Immunodeficiency Virus (HIV) and Importance of Dental Surgery

Aetiology (in Dental Clinic)

- **Infected instruments**
- **Use of infected needles**
- **Infected blood transfusion** during dental procedures
- **Contaminated gloves and dressing materials**

Epidemiology

- **AIDS was first described in the US** and has spread globally.
- **Africa** has **50% of all positive cases** globally.
- **1 in every 100 sexually active adults** worldwide is infected with **HIV**.

Pathology

- Depletion of **CD4+ T cells** leading to:
    - **Selective tropism** and internalization of HIV.

- **Uncoating** and **proviral DNA integration**.
- **Budding and syncytia formation** (cell fusion).
- **Cytopathic effects** (damage to cells).
- Effects on **monocytes and macrophages**.
- **HIV infection of the nervous system**.
- **B cell dysfunction** (affects immune response).

Precautions in Dental Clinic

- **Avoid sharing needles**.
- **Sterilize instruments** properly.
- **ELISA testing** before blood transfusion.
- **Aseptic measures** should be followed before surgery.
- **Educate the patient about AIDS**.